ON A VECTOR SPACE OF DIMENSION BELOW THE CARDINALITY OF THE FIELD SOME MONIC POLYNOMIAL IN ANY OPERATOR IS NOT INVERTIBLE

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Allowed standard prerequisites (editorial).

- The vector space structure on V over the rational function field k(t). Built inside the counted proof at lines 7099-7100. Turns the operator into a scalar and makes V a vector space over the larger field.
- The dimension of k(t) over k is at least the cardinality of k. Used at the end of the counted proof (lines 7100-7102). Supplies the lower bound that contradicts the assumed cardinal bound on the dimension of V.

Lemma 0.1. *Suppose that k is a field and suppose that V is a nonzero vector space over k. Assume the dimension of V (which is a cardinal number) is smaller than the cardinality of k. Then for any linear operator $T : V \to V$ there exists some monic polynomial $P(t) \in k[t]$ such that $P(T)$ is not invertible.*

Proof. If not then V inherits the structure of a vector space over the field $k(t)$. But the dimension of $k(t)$ over k is at least the cardinality of k for example due to the fact that the elements $\frac{1}{t-\lambda}$ are k-linearly independent. □

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